

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Easy

Question

A right triangle has legs with lengths of **11** centimeters and **9** centimeters. What is the length of this triangle's hypotenuse, in centimeters?

Answer

- A. $\sqrt{40}$
- B. $\sqrt{202}$
- C. 20
- D. 202

Correct Answer: B

Rationale

Choice B is correct. The Pythagorean theorem states that for a right triangle, $c^2 = a^2 + b^2$, where c represents the length of the hypotenuse and a and b represent the lengths of the legs. It's given that a right triangle has legs with lengths of **11** centimeters and **9** centimeters. Substituting **11** for a and **9** for b in the formula $c^2 = a^2 + b^2$ yields $c^2 = 11^2 + 9^2$, which is equivalent to $c^2 = 121 + 81$, or $c^2 = 202$. Taking the square root of each side of this equation yields $c = \pm\sqrt{202}$. Since c represents a length, c must be positive. Therefore, the length of the triangle's hypotenuse, in centimeters, is $\sqrt{202}$.

Choice A is incorrect. This is the result of solving the equation $c^2 = 11(2) + 9(2)$, not $c^2 = 11^2 + 9^2$.

Choice C is incorrect. This is the result of solving the equation $c(2) = 11(2) + 9(2)$, not $c^2 = 11^2 + 9^2$.

Choice D is incorrect. This is the result of solving the equation $c = 11^2 + 9^2$, not $c^2 = 11^2 + 9^2$.